

# Customer Segmentation Using RFM and Behavioral Features

Machine Learning Course Project

Vivek Solanki

## Abstract

This project performs customer segmentation on the Online Retail transaction dataset using unsupervised learning. After cleaning raw transactions (removing invalid quantities/prices and creating a total spend feature), customer-level features were engineered including RFM (Recency, Frequency, Monetary) and additional behavioral metrics such as total quantity purchased, unique products bought, average unit price, active days, and number of countries purchased from. All features were standardized and clustering experiments were performed using K-Means and hierarchical clustering. The resulting clusters were profiled and interpreted as meaningful customer segments that can support targeted marketing and customer retention strategies.

## 1 Introduction

Customer segmentation is a common business analytics task where customers are grouped based on purchasing behavior so that marketing actions can be personalized. Instead of treating all customers identically, segmentation can help identify high-value loyal customers, mid-value regular customers, and low-value/infrequent customers. In this project, the machine learning problem is **unsupervised clustering**: given customer-level features derived from transaction history, automatically discover natural groups of similar customers. The dataset used is a public online retail transaction dataset containing invoice-level purchases, product information, customer IDs, and countries.

## 2 Methodology

### 2.1 Dataset

The dataset consists of retail transactions with the following key columns: `InvoiceNo`, `StockCode`, `Description`, `Quantity`, `InvoiceDate`, `UnitPrice`, `CustomerID`, and `Country`. The analysis was implemented in Python using standard data science libraries (pandas, NumPy, scikit-learn, SciPy, and matplotlib).

### 2.2 Data Preprocessing

The raw dataset contained missing values and invalid transaction rows. The following preprocessing steps were applied:

- Removed rows with non-positive quantity (`Quantity`  $\leq 0$ ).

- Removed rows with non-positive unit price ( $\text{UnitPrice} \leq 0$ ).
- Converted `InvoiceDate` to a proper datetime type for time-based calculations.
- Created a new feature  $\text{TotalPrice} = \text{Quantity} \times \text{UnitPrice}$ .

Cancellations (invoices whose `InvoiceNo` begins with ‘‘C’’) were handled separately by computing cancellation-related features from the original dataset while clustering was performed on cleaned positive transactions.

### 2.3 Feature Engineering (Customer-Level Features)

To perform clustering at the customer level, transactions were aggregated by `CustomerID`. A snapshot date was defined as one day after the last invoice date. The following features were constructed:

- **Recency**: days since the customer’s last purchase.
- **Frequency**: number of invoices (purchase count).
- **Monetary**: total spending (sum of `TotalPrice`).
- **TotalQuantityPurchased**: total quantity purchased across all transactions.
- **UniqueProductsBought**: number of unique products (`StockCode`) purchased.
- **AvgUnitPrice**: average unit price of items purchased.
- **ActiveDays**: days between the first and last purchase (customer “lifetime” in days).
- **CountryCount**: number of unique countries associated with the customer’s purchases.
- **CancellationCount**: number of cancellation invoices (from original data).
- **CancellationRate**:  $\text{CancellationCount} / \text{Frequency}$ .

Missing values were filled with zero when needed (e.g., customers with no cancellations). The final customer-level feature table was exported as `Customer_Level_Features.csv`.

### 2.4 Exploratory Data Analysis (EDA)

EDA was performed at both the transaction level and customer level. Key visualizations included the distribution of item quantities (including the **top 10 most frequent quantity values**), identification of the **top 10 highest-value transactions** by `TotalPrice`, the **top 10 countries** by number of orders and by total revenue, the **top 10 most sold products** (by purchase count), and the **top 10 customers** by total spending. A monthly sales trend was also computed by grouping revenue by year-month. These results provide intuition about purchasing patterns, customer concentration, and seasonality, and they motivate feature scaling and careful cluster interpretation.

## 2.5 Clustering Experiments

Because clustering is distance-based, all numeric customer features were standardized using `StandardScaler`. Two clustering approaches were tested:

- **K-Means:** multiple values of  $k$  were evaluated using the elbow method (inertia). Final clustering was explored using  $k = 2$  and  $k = 3$  for interpretability.
- **Hierarchical Clustering:** Ward linkage was used and a dendrogram was created to examine cluster structure and select a cut level.

Two hierarchical clustering experiments were performed. First, hierarchical clustering was applied using all available engineered customer features to understand the overall structure of the customer space. Second, hierarchical clustering was repeated using a selected subset of features to focus on the most informative dimensions for segmentation and interpretation. This helped compare whether the cluster structure remains consistent when using fewer, more meaningful features.

Cluster separability was visualized using Principal Component Analysis (PCA) in 2D (and optionally 3D) by projecting standardized features into a low-dimensional space. During early experiments, clusters were not clearly separated, so additional research was conducted on clustering best practices (feature scaling, choosing  $k$ , outlier handling, and interpretation) before finalizing the approach.

Table 1: Approximate time breakdown (total  $\approx 25$  hours).

| Activity                                                                                                            | Hours |
|---------------------------------------------------------------------------------------------------------------------|-------|
| Learning / refreshing Python for data analysis (pandas, NumPy, datetime handling)                                   | 6     |
| Data loading, cleaning decisions, and creating the cleaned dataset (TotalPrice, filtering invalid rows)             | 5     |
| Exploratory Data Analysis (distributions, top countries/products/customers, monthly trend)                          | 4     |
| Customer-level feature engineering (RFM + behavioral features such as ActiveDays, CountryCount, etc.)               | 4     |
| Clustering experiments and troubleshooting (scaling, PCA visualization, comparing $k$ , interpreting clusters)      | 5     |
| Research to improve clustering quality when clusters were not clearly separated (reading docs, trying alternatives) | 1     |
| Final report writing, LaTeX formatting, and assembling figures/tables                                               | 0.5   |

The time above includes roughly **24 hours** of learning, experimentation, and implementation work, plus additional time for final report formatting.

Table 2: Comparison of clustering solutions. Replace metric values with your computed results from Python.

| Method              | k                    | Sil $\uparrow$  | DB $\downarrow$ | CH $\uparrow$   | Interpretability / Notes                                        |
|---------------------|----------------------|-----------------|-----------------|-----------------|-----------------------------------------------------------------|
| K-Means             | 2.00 ( <i>fill</i> ) | ( <i>fill</i> ) | ( <i>fill</i> ) | ( <i>fill</i> ) | Very simple split; may merge mid- and high-value behaviors.     |
| K-Means             | 3.00 ( <i>fill</i> ) | ( <i>fill</i> ) | ( <i>fill</i> ) | ( <i>fill</i> ) | More actionable segments; clearer profiles across features.     |
| Hierarchical (Ward) | 3.00 ( <i>fill</i> ) | ( <i>fill</i> ) | ( <i>fill</i> ) | ( <i>fill</i> ) | Dendrogram explains structure; useful to justify the cut level. |

**Comparison criteria.** To compare K-Means ( $k = 3$ ) with hierarchical clustering (Ward,  $k = 3$ ), we used two complementary checks:

1. **Cluster quality (internal validation):** we compared Silhouette, Davies–Bouldin, and Calinski–Harabasz scores (Table 2).
2. **Stability / agreement:** we measured how similar the two labelings are using an agreement metric such as the Adjusted Rand Index (ARI), where a higher value indicates stronger agreement.

If both methods achieve similar validation scores and produce consistent groupings, it increases confidence that the segments reflect real structure in the data rather than an artifact of a single algorithm.

**Final choice.** In this project, the final reported segmentation uses **K-Means with  $k = 3$**  because it produced segments that were easier to interpret and describe using the engineered features, while maintaining competitive internal validation scores. Hierarchical clustering was used as a **confirmatory method** to validate that a similar 3-group structure exists and to visualize the nested structure via the dendrogram.

### 3 Conclusion

This project transformed raw retail transaction data into customer-level behavioral features and applied unsupervised clustering to discover distinct customer segments. The final clusters show meaningful differences in spending, purchase frequency, and engagement, enabling actionable strategies such as loyalty programs for high-value customers and re-engagement campaigns for infrequent customers. A limitation is that clustering can be sensitive to outliers and feature scaling; future work could explore robust preprocessing and additional models such as Gaussian Mixture Models (GMM) or density-based clustering, and could evaluate segment stability over time.

## References

## References

- [1] scikit-learn documentation: K-Means clustering. <https://scikit-learn.org/stable/modules/clustering.html#k-means>
- [2] SciPy documentation: Hierarchical clustering (`scipy.cluster.hierarchy`). <https://docs.scipy.org/doc/scipy/reference/cluster.hierarchy.html>
- [3] Online Retail dataset (public dataset source; replace with your exact source link). <https://archive.ics.uci.edu/>
- [4] OpenAI ChatGPT (GPT-5.2) assistance for code structuring, clustering interpretation, and LaTeX report formatting (Accessed: Dec. 2025). <https://chat.openai.com/>